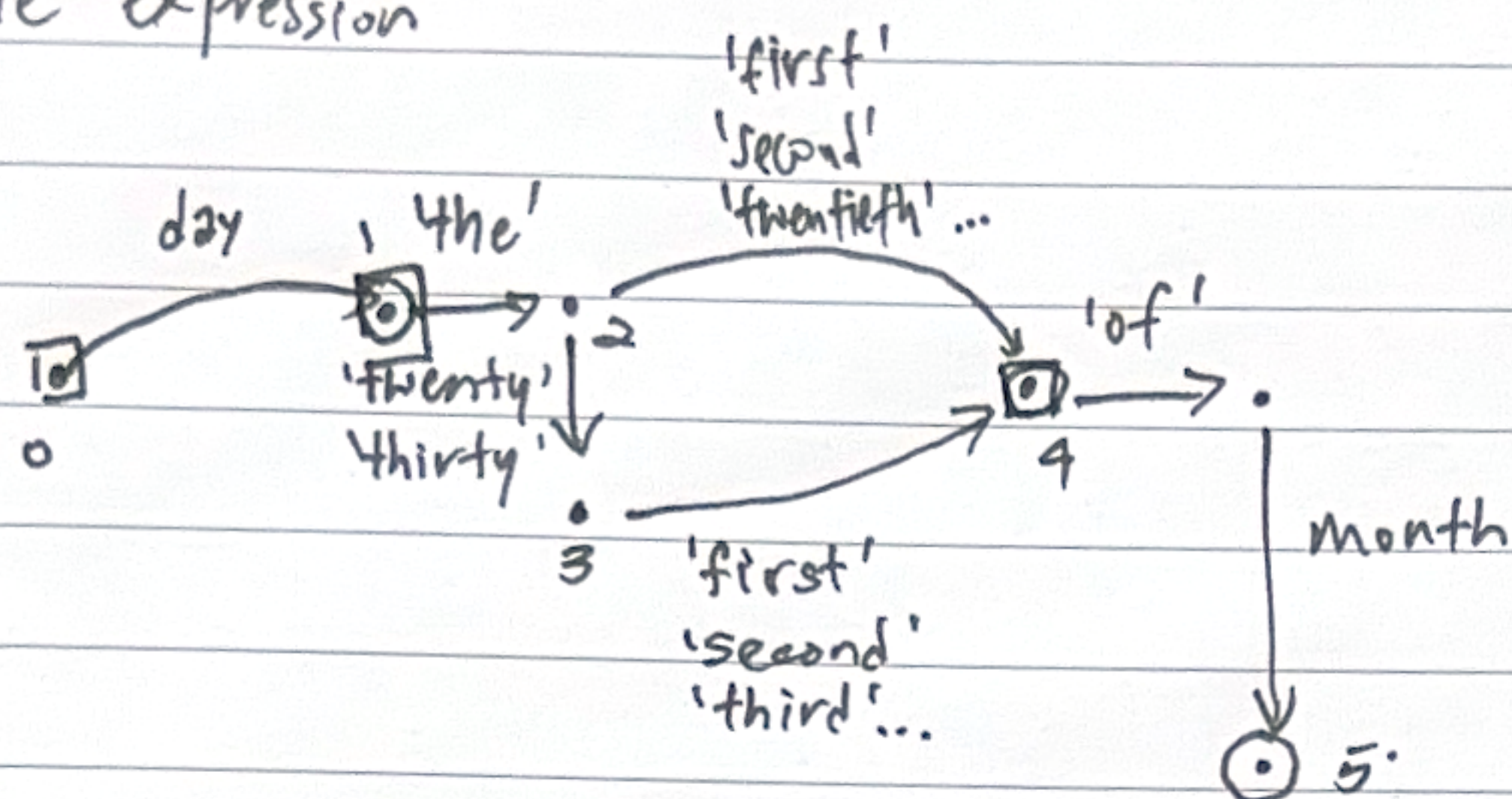


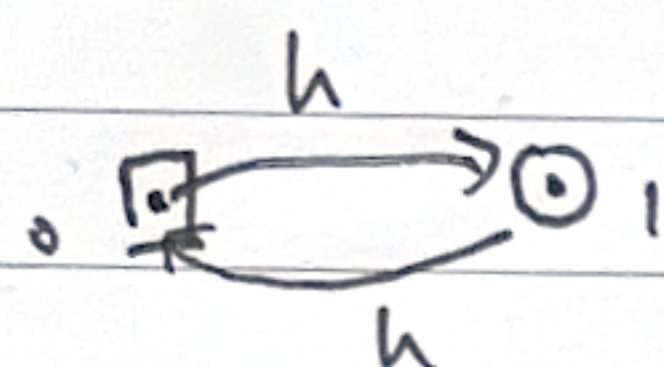
Q1. Date Expression



start: 0, 1

end: 1, 4, 5.

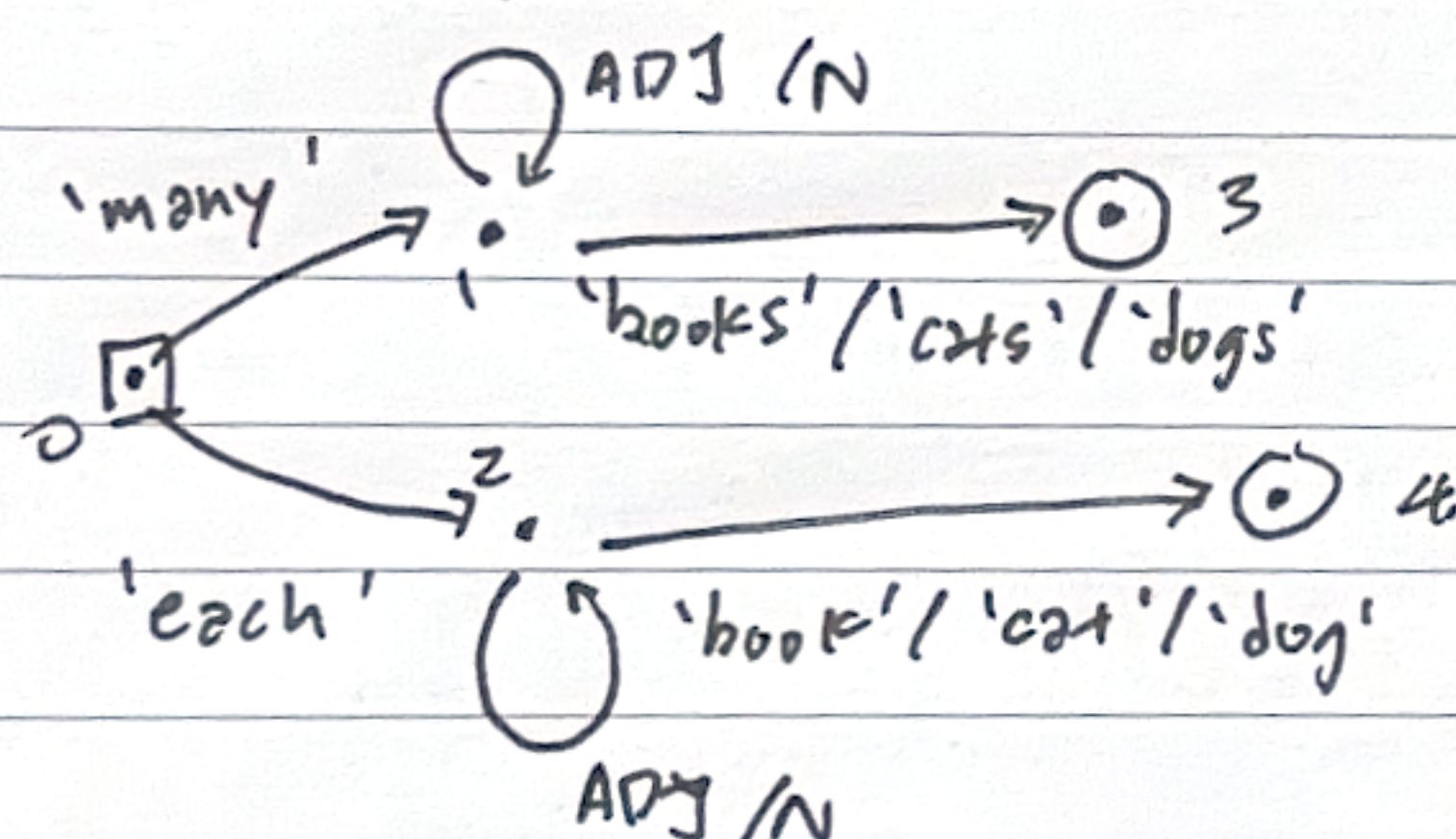
Q2. Odd.



start: 0

end: 1

Q3. Long Distance Agreement



start: 0

end: 3, 4

Q4. Pumping Lemma

Theorem 1: $xy^n z \in L; n \geq 0$.

$$L = \{a^n i (bc)^n \mid n \geq 0\}$$

if L is a regular language, then $xy^n z \in L$.

1. $y = a$: will lead to an imbalanced no. of a compared to bc . $\notin L$.

2. $y = bc$: will lead to an imbalanced no. of $'bc'$ eq. than $'a'$. Also breaks L .

3. $y = i$: such that $\underbrace{(a^x)}_x \underbrace{(i^y)}_y \underbrace{(bc)^z}_z$. However, $n \geq 0$ for regular languages. For L to be valid, $n \neq 0$.

Thus, since there cannot be a FSA to recognize $L = \{a^n i (bc)^n \mid n \geq 0\}$